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Handling Multiple Windows and Frames

MARCH, 2025



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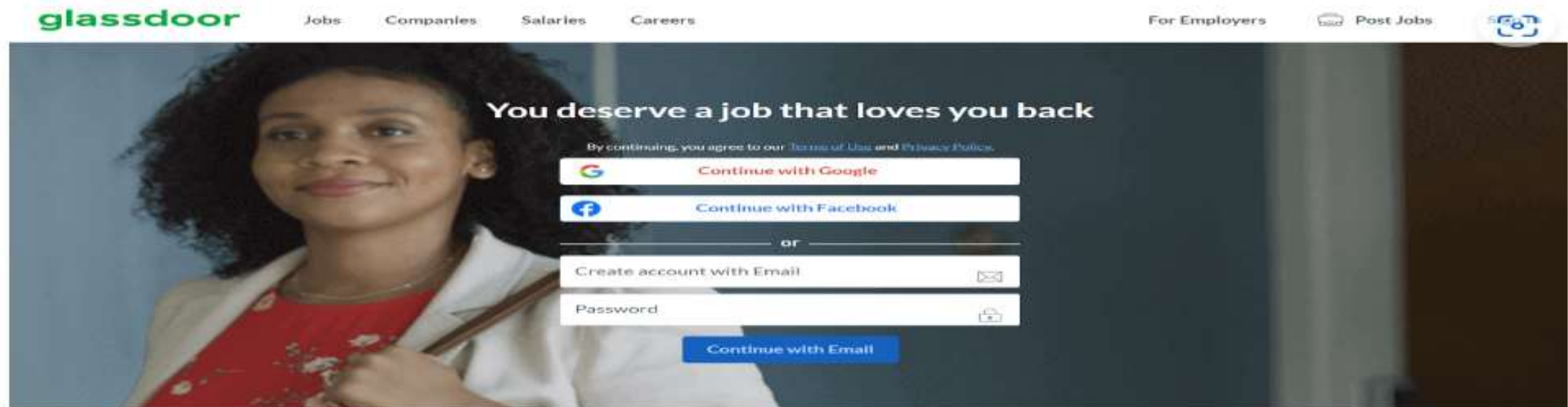
Windows in Selenium



Windows in Selenium

Introduction

- In Selenium Automation Testing, when we talk about “windows”, we’re referring to the different browser windows or tabs that can open up when you interact with a web application.
- These windows can be the primary browser window or ones that arise after clicking links or buttons.
- For example, the job search platform Glassdoor opens a new window when you click the buttons to sign in using your Google account or Facebook account.

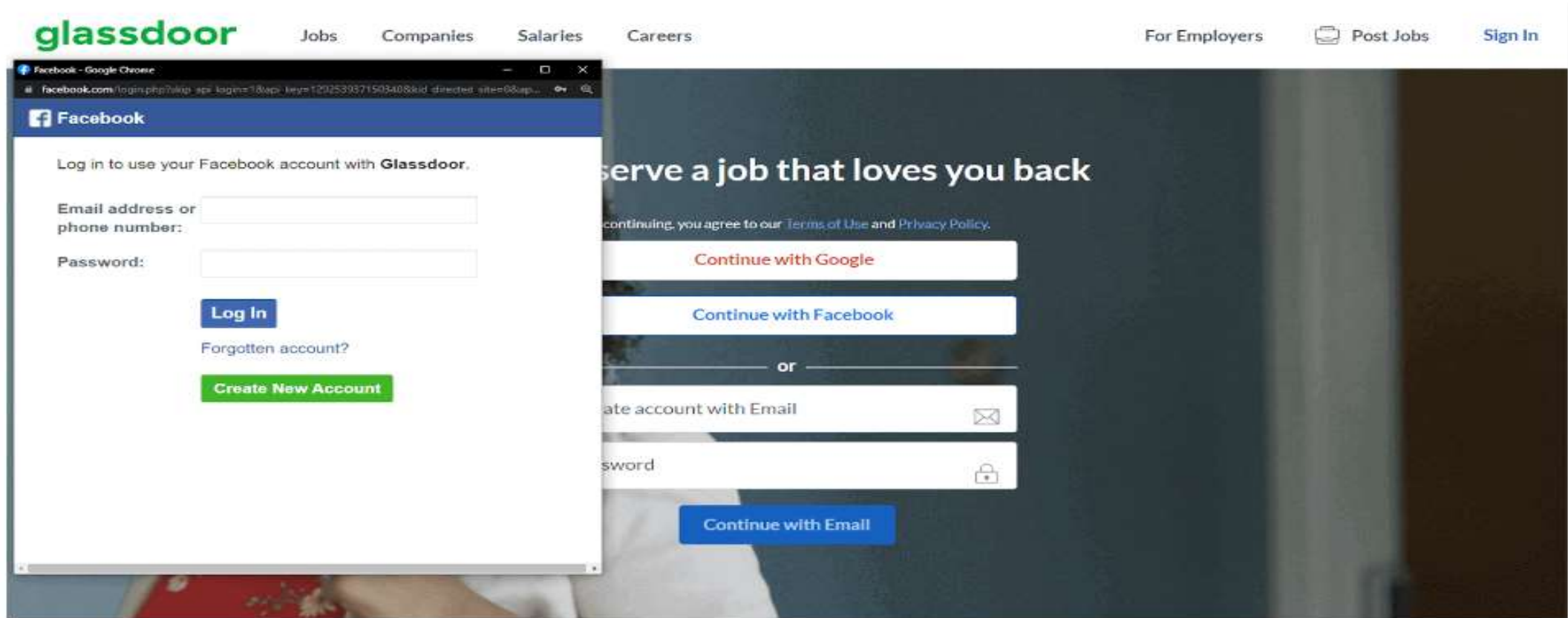


Glassdoor Home Page

Windows in Selenium

Introduction

- Click on the “Continue with Facebook” button to sign in.



Facebook Page Window

Windows in Selenium

Introduction

- A Facebook webpage window will pop up to prompt you to sign in.
- As a result, the user might need to access the open browser window to move forward.
- As a QA tester, running a test case might require you to complete some operations in a newly opened browser window or tab before going back to the original window or tab to finish the rest of the tasks.
- There are two types of Windows In Selenium
 1. Main Window
 2. Child Window

Windows in Selenium

Introduction

1. Main Window

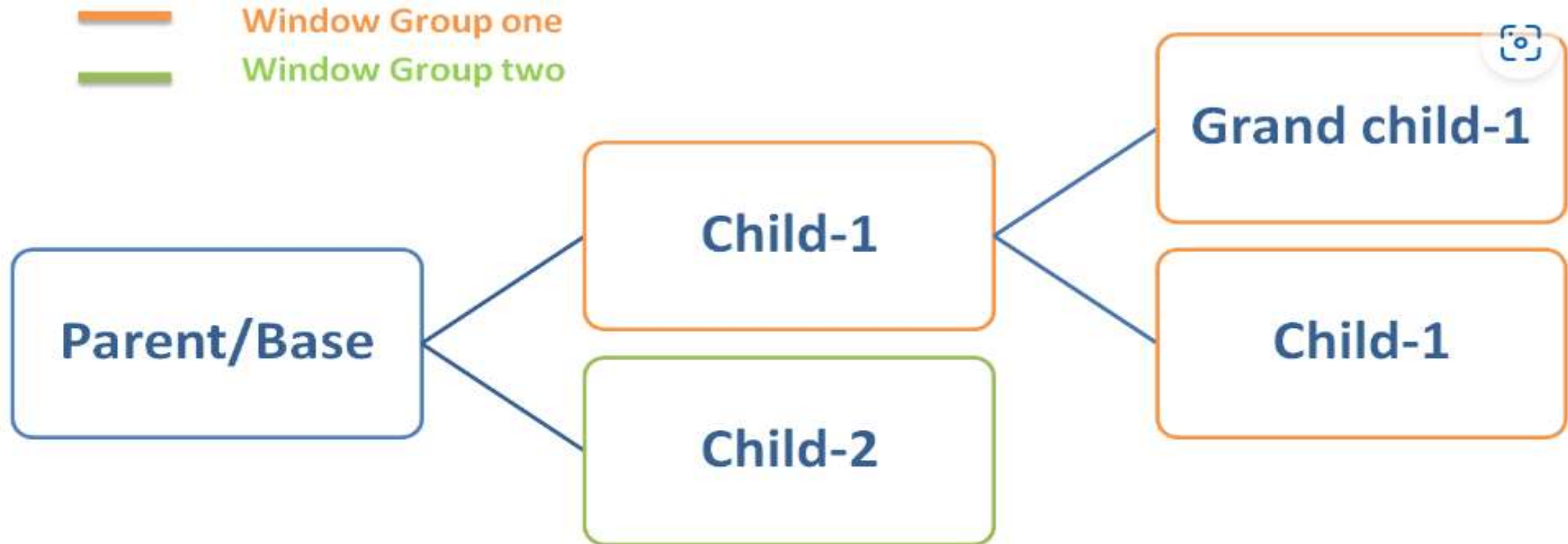
- The main window is the initial browser window that opens when you start a Selenium test. It's like the front door of a house.
- You interact with the elements on this window as you navigate through the web application.
- **For Example:**
- Let's say you're testing an online shopping website.
- The main window is where you land when you open the website.
- You can click on different products, add them to your cart, and navigate through various pages – all within this main window.

Introduction

2. Child Window

- Child windows, on the other hand, are additional windows that pop up when you perform certain actions.
- These could be like opening new doors within the house that lead to different rooms.
- **For Example:**
- Back to the online shopping example, imagine you're looking at a product and you click a "Details" button.
- A new window might pop up with more information about the product. This is a child window.
- You can interact with elements in this window, like reading the product description, without losing track of the main window.

Introduction



The Hierarchy of Browser Windows

Windows Handle and Methods



Windows Handle and Methods

Window Handle

- Multiple windows are brought up by scenarios in which a web application navigates to or opens multiple browser windows.
- Selenium provides the option to handle multiple windows with window handles.
- A **window handle** is like an **address for browser windows**.
- It's a **unique code** made of letters and numbers that **helps Selenium identify and manage windows**.
- Just as a key opens a specific room in a building, a window handle lets Selenium switch between different windows.
- Each window has its own handle, making it easy to work with multiple windows and their child windows while testing a website..

Methods To Handle Windows In Selenium

- 1) **GetWindowHandle():** This method returns the unique handle of the currently focused window. It's often used to identify the main or parent window.

```
| String parentHandle= driver.getWindowHandle();
```


Windows Handle and Methods

Methods To Handle Windows In Selenium

2) GetWindowHandles(): This method retrieves the handles of all open windows. It returns a set of strings, each representing a unique window handle.

```
Set handle= driver.getWindowHandles();
```

Methods To Handle Windows In Selenium

3) SwitchTo Window: This method allows you to switch your focus between different windows. You provide the window handle as an argument, and Selenium switches your interaction context to that window.

```
driver.switchTo().window("windowName");
```

How to handle multiple windows



How to handle multiple windows

step by step guide on how to handle multiple windows

Step 1: Open Main Window

Step 2: Trigger Action for New Window

Step 3: Capture Window Handles

Step 4: Switch to New Window

Step 5: Do Stuff in the New Window

Step 6: Switch Back to Main Window

Step 7: Continue Testing on Main Window

Step 8: Repeat as Needed

Step 9: Close or Quit

How to handle multiple windows

Example Scenario

Let us explain window switch with an example:-

- 1) Launch new Browser and open <https://demoqa.com/browser-windows>
- 2) Check the count of windows which is 1
- 3) Locate “New Window” button using Id – “windowButton” and click to open a new window
- 4) Get the count of both windows which is now 2.
- 5) Get the parent window handle and print it to console
- 6) Get the window handles of both the open windows and print them
- 7) Switch to the new window (child window)
- 8) Get the text of Child Window and print it
- 9) Close the new window (child window)

Handling Multiple Windows - Demo

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import java.util.Iterator;
import java.util.Set;
import java.util.concurrent.TimeUnit;

public class WindowSwitchDemo {

    public static void main(String[] args) {

        ChromeOptions options = new ChromeOptions();
        options.addArguments("--start-maximized");
        WebDriver driver = new ChromeDriver(options);
        driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);

        // Pass application url
        driver.get("https://demoqa.com/browser-windows");
        String parentwindow = driver.getWindowhandle();
        SOP(parentwindow);
```


Handling Multiple Windows - Demo

```
//Count of window - 1
Set<String> allWindowHandles = driver.getWindowHandles();
System.out.println("Count of Window :" + allWindowHandles.size());

//Open a child window
driver.findElement(By.id("windowButton")).click();

//Count of windows , changed from 1 to 2
Set<String> newAllWindowsHandles = driver.getWindowHandles();
System.out.println("New Count of Window :" + newAllWindowsHandles.size());

// Get the detail of the parent window
String ParentHandle = driver.getWindowHandle();
System.out.println("Parent Window :" + ParentHandle);

//Get details of parent and child windows
Iterator<String> iterator = newAllWindowsHandles.iterator();
String mainWindow = iterator.next();
String childWindow = iterator.next();
System.out.println("Parent Window :" + mainWindow);
System.out.println("Child Window :" + childWindow);
```

Handling Multiple Windows - Demo

```
//Switch control to child window
driver.switchTo().window(childWindow);

//Verify the text present on child window
WebElement text = driver.findElement(By.id("sampleHeading"));
System.out.println("Child_Title :" + text.getText());

// Close Child window
driver.close();

// Switch back to parent window
driver.switchTo().window(ParentHandle);
System.out.println("Parent Title :" + driver.getTitle());

// Close Parent window
driver.quit();
}
}
```

Output

Output:

```
Run: WindowSwitchDemo x
"C:\Program Files\jdk-17_windows-x64_bin\jdk-17.0.9\bin\java.exe" ...
Count of Window :1
New Count of Window :2
Parent Window :0AF534534EAB881B6977A681B3487566
Parent Window :0AF534534EAB881B6977A681B3487566
Child Window :DF52144F74459BFD577C1DDF10818CF7
Child_Title :This is a sample page
Parent Title :DEMOQA

Process finished with exit code 0
```

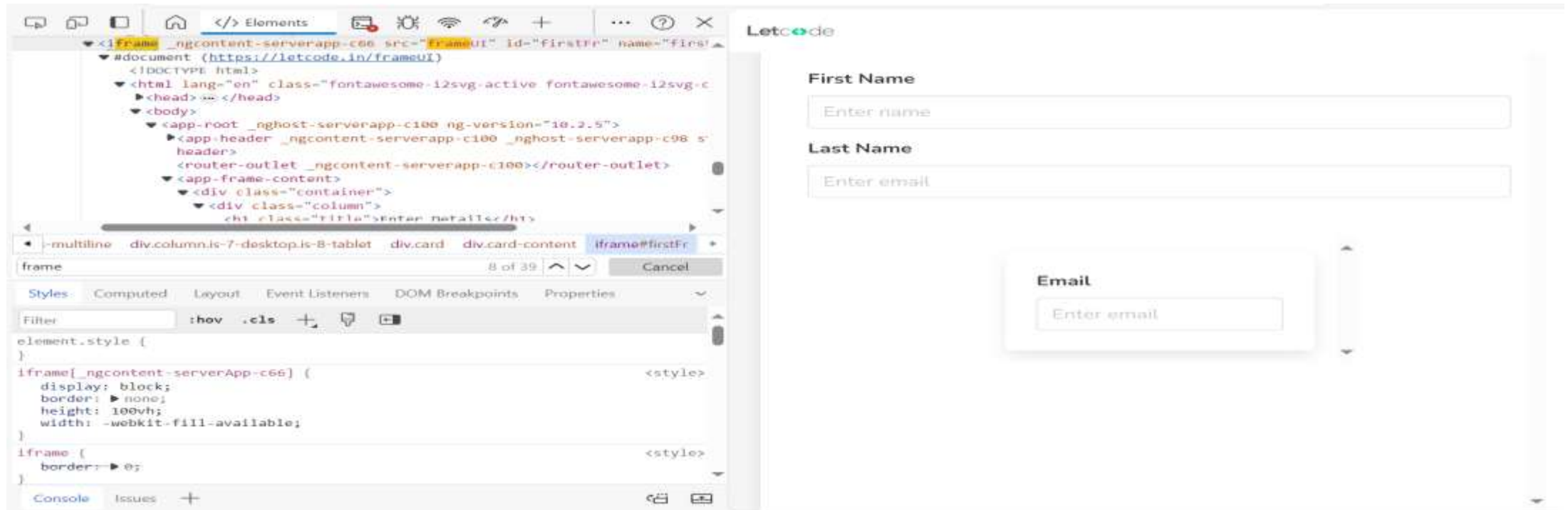
Handling Frames



Handling Frames

iframes

- An iframe is also known as an inline frame. It is a tag used in [HTML5](#) to embed an HTML document within a parent HTML document.
- An iframe tag is defined using `<iframe></iframe>` tags



Handling Frames

frame vs iframe

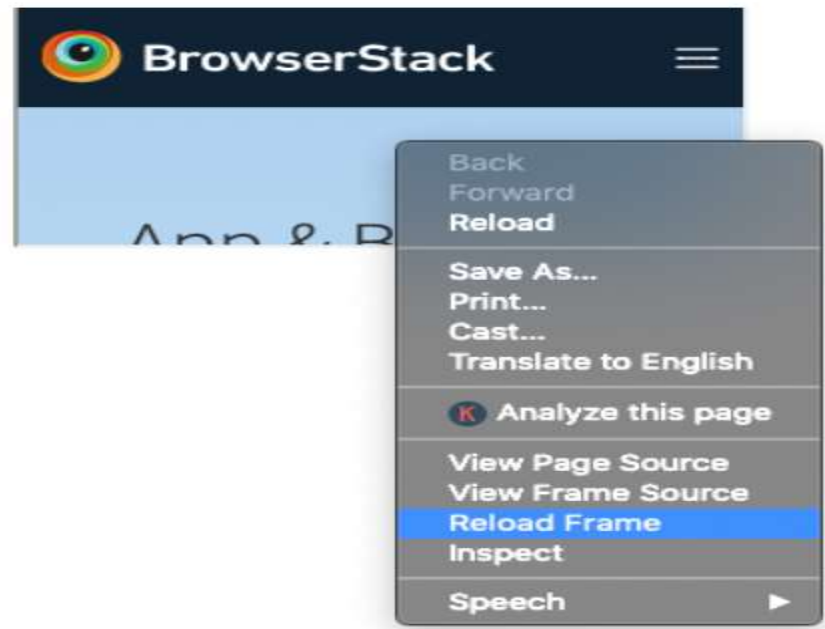
- The **frame** enables a developer to split the screen horizontally or vertically by using the frameset tag.
- **Note:** Frame and frameset tags are deprecated as they are no longer supported by HTML 5.
- The iframes are mainly used to insert content from external sources. For example, an advertisement displayed on a web page. They can float within the webpage, which means one can position an iframe at a specific position on a web page.



Handling Frames

How to identify a frame on a page?

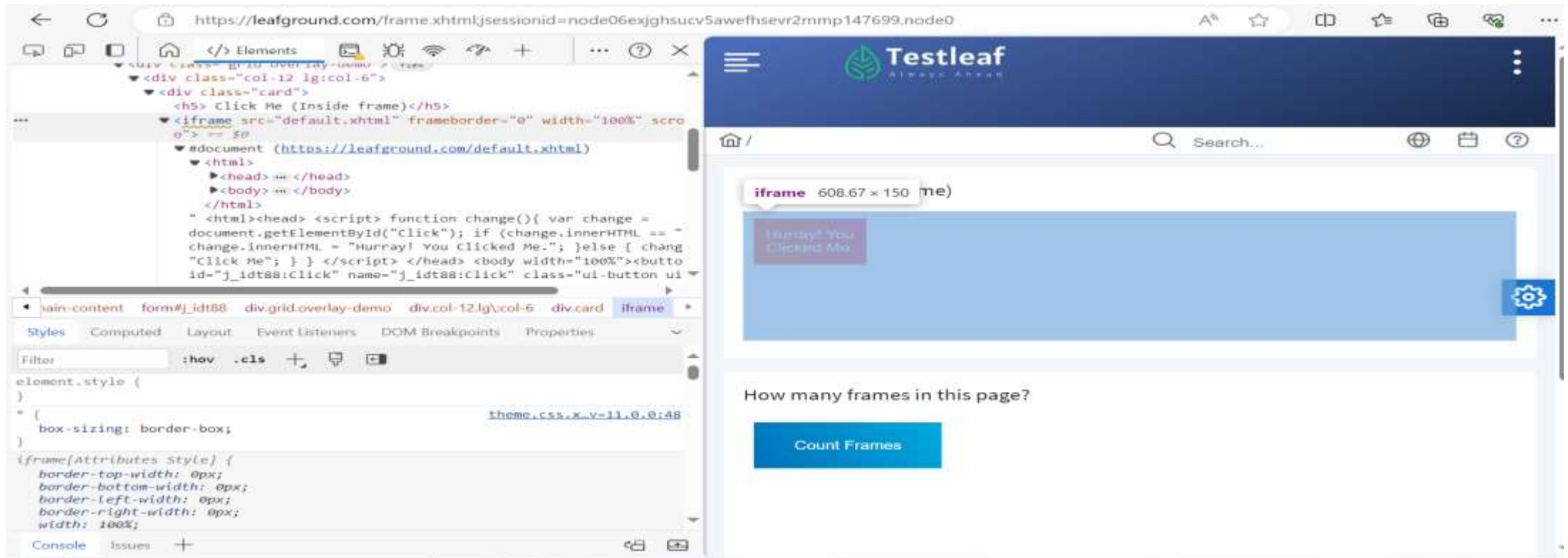
- It is possible to identify the iframes on a web page in two ways:
- Right-click on the specific element and check all the options. If you find an option like **This Frame, view Frame source or Reload Frame**, the page includes frames.
- Consider the image below as an example.



Handling Frames

How to identify a frame on a page?

- Similar to the first step, right-click on the page and click on View Page Source. On the page source, **search for “iframe-tags”**. If you find any iframe tags, it means the page includes iframes.



Handling Frames

Using the `switchTo().frame`

- This method enables the browser to switch between multiple frames. It can be implemented in the following ways:

1. Switching by Name or ID

- If the frame or iframe has an id or name attribute, we can switch the frames using name or ID. If the name or ID is not unique on the page, then the first one found will be switched to.

Example:

```
//switch To IFrame using name or id
driver.findElement(By.name("iframe1-name"));

//Switch to the frame
driver.switchTo().frame(iframe);
```

Handling Frames

Using the `switchTo().frame`

- This method enables the browser to switch between multiple frames. It can be implemented in the following ways:

2. Switching by WebElement

- We can find the frame using any selector and switch to it.

Example:

```
WebElement frameElement = driver.findElement(By.id("frameId"));  
driver.switchTo().frame(frameElement);
```

Handling Frames

Using the `switchTo().frame`

- This method enables the browser to switch between multiple frames. It can be implemented in the following ways:

3. Switching by Index

- Switching between the frames can be done by Index also.

Example:

```
//switch To IFrame using index  
driver.switchTo().frame(0);
```

Handling Frames

Using the `switchTo().frame`

- This method enables the browser to switch between multiple frames. It can be implemented in the following ways:

4. Switching back to the default content:

- To leave an iframe or frameset, switch back to the default content like:

Example:

```
//leave frame  
driver.switchTo().defaultContent();
```

Handling Frames

Using the `switch_to().frame` function

Example:

- Suppose there are three frames named i1,i2 and i3 within the parent web page p1. Frames i1, i2, and i3 are dependant on each other which means one frame will be the parent of another.
- Using `driver.switchTo().defaultContent()` method on frame i3, web driver control moves to the parent page, p1. While `driver.switchTo().parentFrame()` method on frame i3 switches the control back to frame i2 and so on.

Handling Frames

Using the `switch_to().frame` function

Example:

Let us explain frame switchching with an example:-

- 1) Launch new Browser and open <https://demoqa.com/frames>
- 2) Switch iFrame using any of locator strategy
- 3) Switch back to main content
- 4) Switch iFrame using index
- 5) Close the window

Handling Frames - Demo

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;

public class iFrame_Demo {

    public static void main(String[] args) {

        ChromeOptions options = new ChromeOptions();
        WebDriver driver = new ChromeDriver(options);
        driver.manage().window().maximize();
        driver.get("https://demoqa.com/frames");

        //Switch iFrame using any of locator strategy
        WebElement iframeElement = driver.findElement(By.id("frame1"));
        driver.switchTo().frame(iframeElement);
        String Frame_1 = driver.findElement(By.id("sampleHeading")).getText();
        System.out.println("Switch by locator:" + Frame_1);
    }
}
```

Handling Frames - Demo

```
//Switch back to the main window
driver.switchTo().defaultContent();
String mainPage = driver.findElement(By.xpath("//*[@id='framesWrapper']/h1")).getText();
System.out.println("Back to Main page :" + mainPage);
```

```
//Switch iFrame using index
driver.switchTo().frame(1);
String Frame_2 = driver.findElement(By.id("sampleHeading")).getText();
System.out.println("Switch by Index :" + Frame_2);
```

```
//quit the browser
driver.quit();
```

```
}
}
```

Handling Multiple Windows - Demo

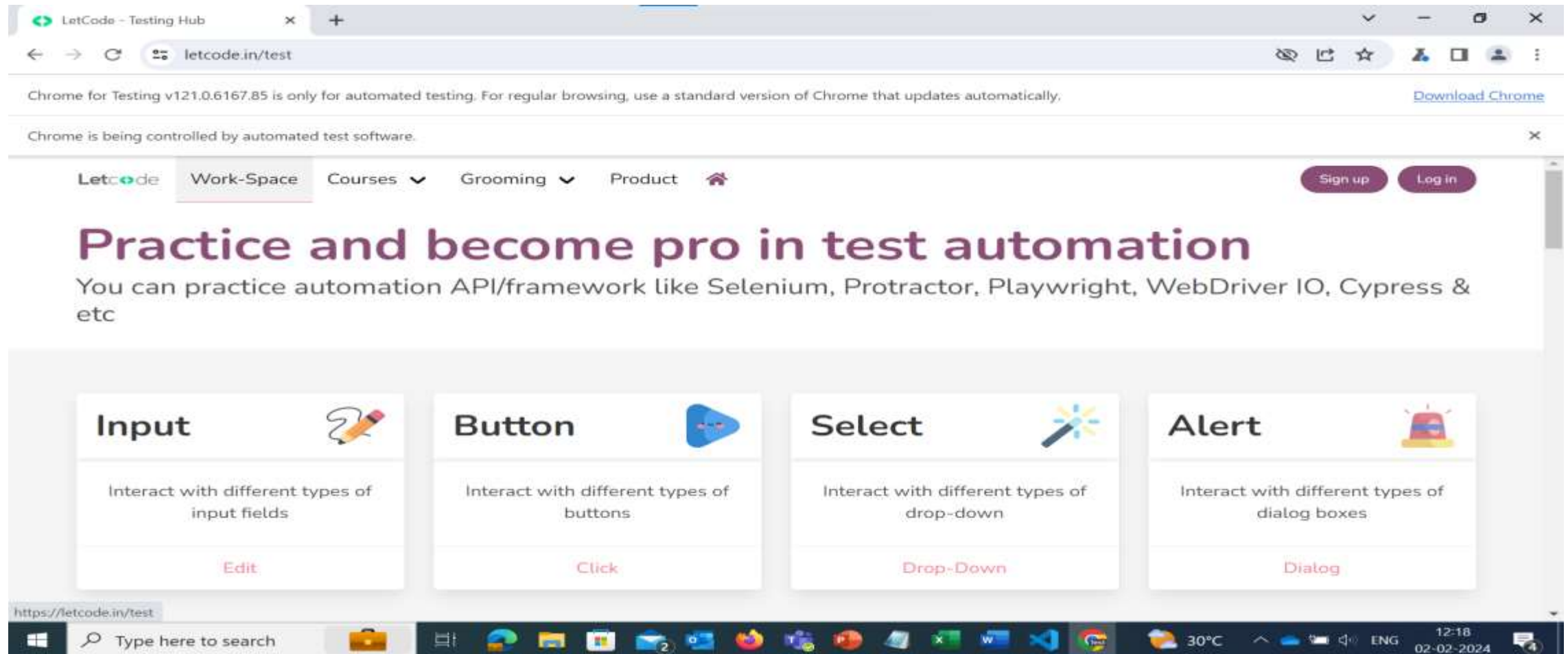
The output of the above program is

```
Switch by locator:This is a sample page  
Back to Main page :Frames  
Switch by Index :This is a sample page
```

Note:-

1. Make sure to always switch back to the default content after performing actions within a frame. This is particularly important before interacting with elements that are outside of the frame.
2. If you want to switch between multiple frames, then switch to a frame and then switch back to main window and then switch to another frame.

Handling Multiple Windows - Demo



THANK YOU